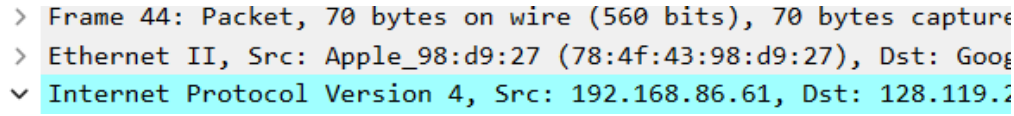
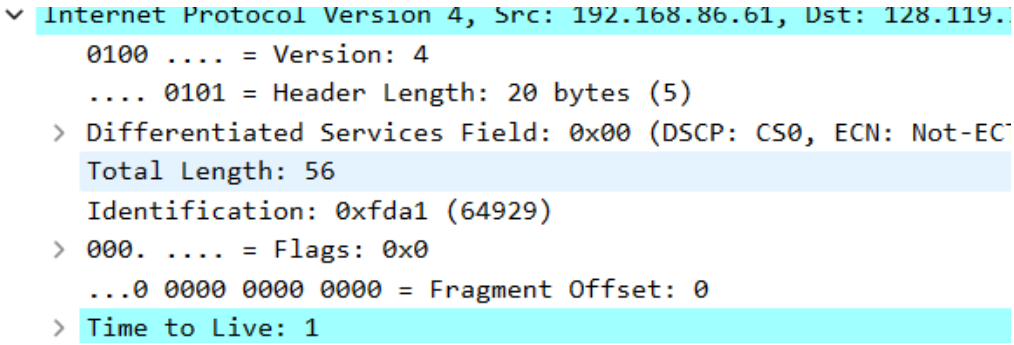
Assignment #6 - Wireshark IP Lab

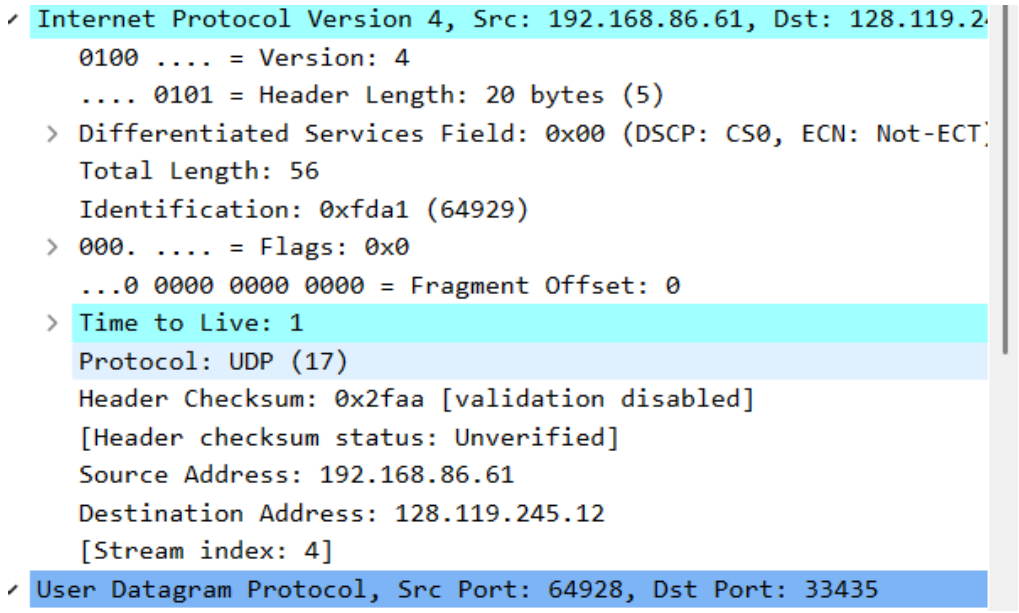
- **Question 1: Select the first UDP segment sent by your computer via the traceroute command to gaia.cs.umass.edu (This is the 44th packet in the trace file in the ip-wireshark-trace1-1.pcapng file). Expand the Internet Protocol part of the packet in the packet details window. What is the IP address of your computer?**
  - IP Address: 192.168.86.61



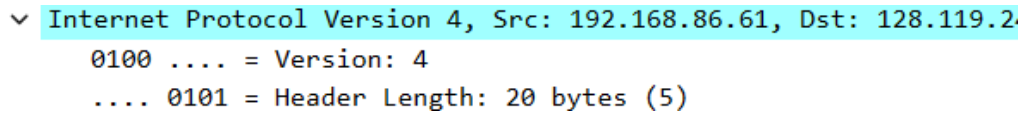
- **Question 2: What is the value in the time-to-live (TTL) field in this IPv4 datagram's header?**
  - TTL Value: 1



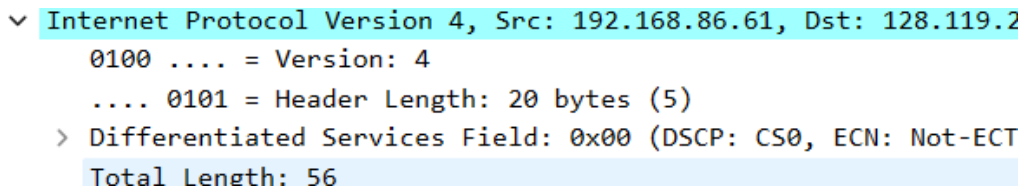
- **Question 3: What is the value in the upper layer protocol field in this IPv4 datagram's header?**
  - Upper Layer Protocol Field Value: UDP, which is represented by "17"



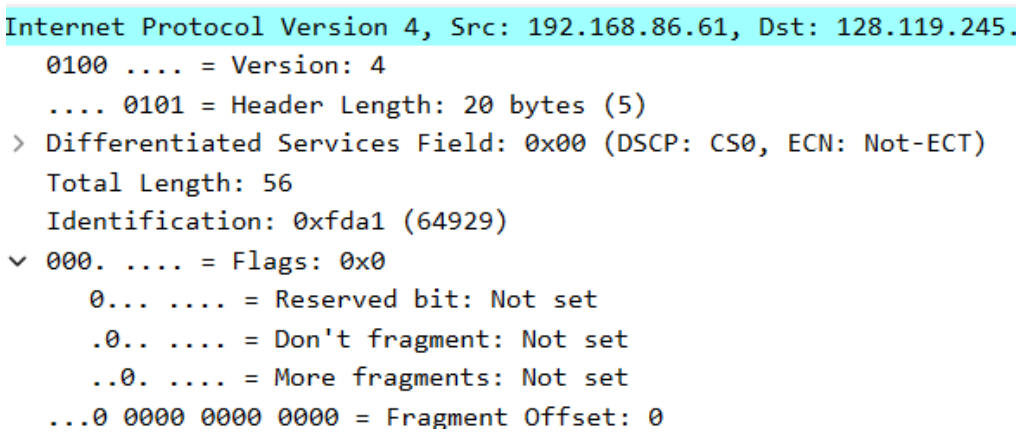
- **Question 4: How many bytes are in the IP header?**
  - Byte Amount: 20 bytes



- **Question 5: How many bytes are in the payload of the IP datagram? Explain how you determined the number of bytes.**
  - Payload Byte Count: 36
  - Total Length = 56, header is 20 bytes, 56 - 20 = 36



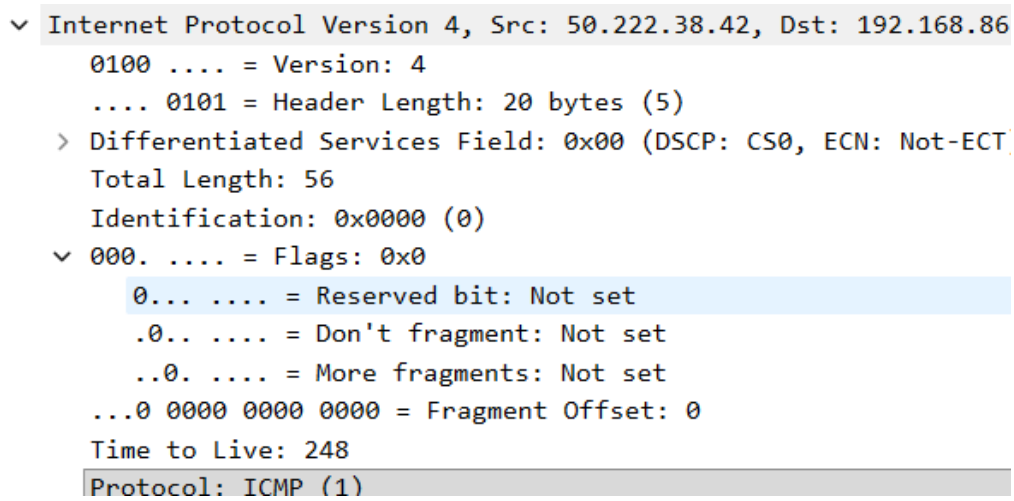
- **Question 6: Has this IP datagram been fragmented? Explain how you determined whether or not the datagram has been fragmented.**
  - No it has not been fragmented
  - The more fragments bit has not been set



- **Question 7: Which fields in the IP datagram always change from one datagram to the next within this series of UDP segments sent by your computer destined to 128.119.245.12, via tracerouter? Why?**
    - The fields that always change in the IP datagram are header checksum, identification number, time to live
    - Header Checksum is always recalculated if something changes
    - Identification Number: A unique number to identify a datagram
    - TTL: If the destination is not reached, the TTL is increased to reach further until the destination is reached. Also goes down by at least 1 from each router.
- **Question 8: Which fields in this sequence of IP datagrams (containing UDP segments) stays constant? Why?**
    - Version: We aren't changing the protocol mid-transit, it must stay the same during the process of traceroute
    - Header Length: We aren't changing anything about the length when traceroute is being run. Only specific fields are changed, but they do not change the length.
    - Total Length: We aren't modifying the header length or the payload length when using traceroute
    - Flags: None of the fragmentation flags are being set or used
    - Protocol: There is no reason to modify the protocol, it just shows what protocol is being used in the encapsulated packet

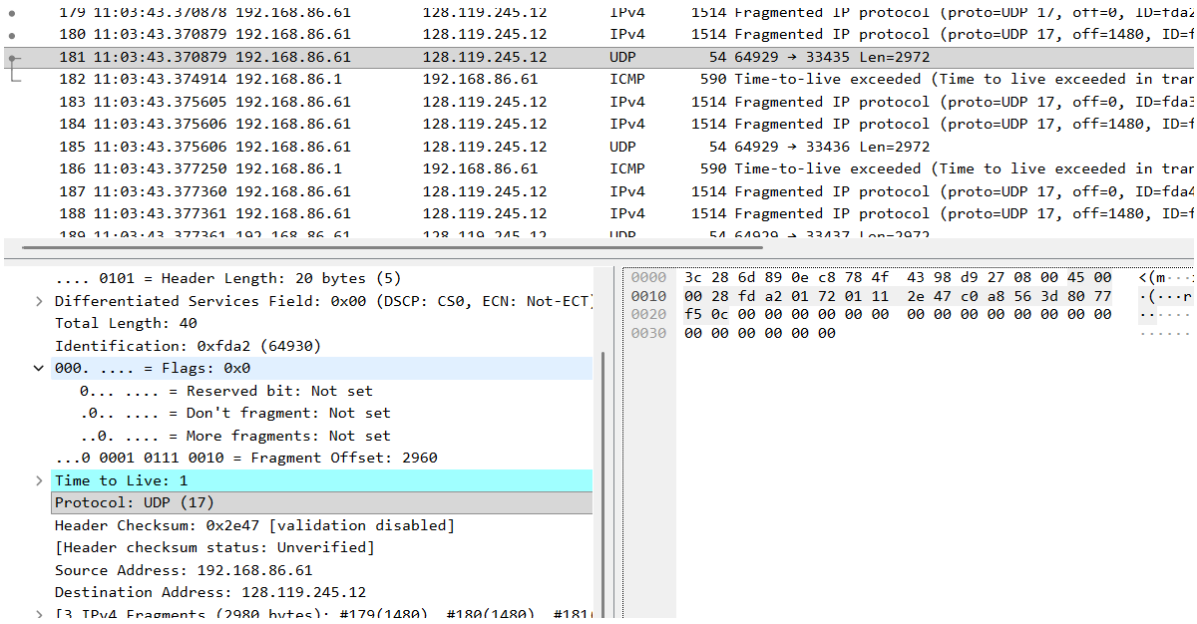
- Source Address: This is where we are sending from and where we want to send a response to. If we modify it, we won't get a response from traceroute
- Destination Address: This is where we are trying to reach, modifying it makes no sense as we are trying to reach this specific address not a different one

- **Question 9: Describe the pattern you see in the values in the Identification field of the IP datagrams being sent by your computer**
  - They increase linearly by 1 for every IP datagram being sent
- **Question 10: What is the upper layer protocol specified in the IP datagrams returned from the routers?**
  - Upper layer protocol: ICMP, represented by a value of "1"

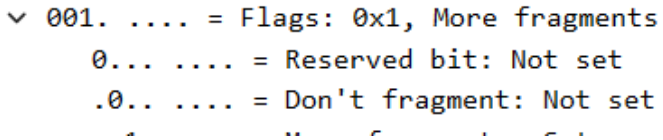


- **Question 11: Are the values in the Identification fields (across the sequence of all of ICMP packets from all of the routers) similar in behavior to your answer to Question 9 above?**
  - Yes, but they are per router instead of a global incremented value.
  - Generally, there they do increase by 1, but as we have only filtered for ICMP, some packets will jump identification numbers by larger amounts (such as 3 or 5), due to background processes and packets consuming the sequential numbers

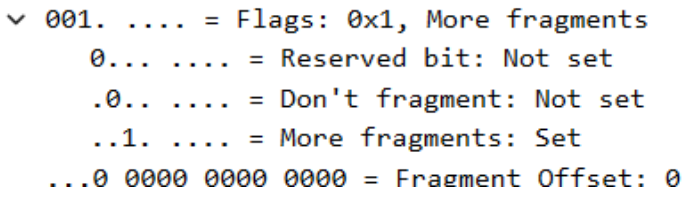
- **Question 12: Are the values of the TTL fields similar, across all of ICMP packets from all of the routers?**
    - No, some of the routers use 250, 64, 55. They are all different, but the same router will decrease the TTL by 1.
- **Question 13: Find the first IP datagram containing the first part of the segment sent to 128.119.245.12 sent by your computer via the traceroute command to gaia.cs.umass.edu, after you specified that the traceroute packet length should 3000. (Hint: this is packet 179 in the ip-wireshark-trace1-1.pcapng trace file. Packets 179, 180, and 181 are three IP datagrams created by fragmenting the first single 3000-byte UDP segment sent to 128.119.145.12). Has that segment been fragmented across more than one IP datagram?**
    - Yes. It has been fragmented into 3 packets



- **Question 14: What information in the IP header indicates that this datagram has been fragmented?**
    - The More Fragments flag has been set, meaning there are more fragments to follow



- **Question 15: What information in the IP header for this packet indicates whether this is the first fragment versus a latter fragment?**
  - The More Fragments flag has been set, indicating that it is at least not the last fragment.
  - The Fragment Offset is also at “0”, which is the start



- **Question 16: How many bytes are there in this IP datagram (header plus payload)?**
  - Fragment 1:
    - 20 (header) + 1480 (payload) = 1500
  - Fragment 2:
    - 20 (header) + 1480 (payload) = 1500
  - Fragment 3:
    - 20 (header) + 40 (payload) = 60
  - Total (after all reassembly) = 3060 bytes
- **Question 17: Now inspect the datagram containing the second fragment of the fragmented UDP segment. What information in the IP header indicates that this is not the first datagram fragment?**
  - The Fragment Offset is greater than 0.

```
001. .... = Flags: 0x1, More fragments
    0... .... = Reserved bit: Not set
    .0.. .... = Don't fragment: Not set
    ..1. .... = More fragments: Set
...0 0000 1011 1001 = Fragment Offset: 1480
```

- **Question 18: What fields change in the IP header between the first and second fragment?**
  - The Fragment Offset and the Header Checksum
- **Question 19: Now find the IP datagram containing the third fragment of the original UDP segment. What information in the IP header indicates that this is the last fragment of that segment?**
  - The More fragment flag is not set, which tells us that this fragment has no more following it.
  - The Fragment Offset is greater than 0, so we know this isn't the first fragment.
  - With both of these two information pieces, we can conclude that the fragment is not the first and cannot be a middle fragment as there are no more fragments after this one.

```
Internet Protocol Version 4, Src: 192.168.86.61, Dst: 128.119.2
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT
  Total Length: 40
  Identification: 0xfda2 (64930)
v 000. .... = Flags: 0x0
    0... .... = Reserved bit: Not set
    .0.. .... = Don't fragment: Not set
    ..0. .... = More fragments: Not set
  ...0 0001 0111 0010 = Fragment Offset: 2960
```